SDBench

We reviewed the GPT-5 baseline trajectory on Microsoft SDBench's synthetic case 0. The final diagnosis was correct (Judge 4/5), but we found four problems in how the agent received information. In two, the Gatekeeper returned results that were not requested. In the other two, it generated results that are not in the source case report. All four come from Gatekeeper responses, not from the diagnostic model. See below.

Audited source · 304ce9625a04 ↗ Run 2026-10-25 · 1 case · findings from the saved run only

Skin biopsy returned the full immunohistochemistry panel `test_logs/53`

Beyond the request | Requested items not reported

Source-backed

The agent ordered deep punch biopsies of the skin lesions with H&E, direct immunofluorescence and bacterial/fungal/AFB stains. It did not name CD20, PAX5, Ki-67 or EBER. The Gatekeeper also returned the source report's immunohistochemistry findings, while answering the immunofluorescence and stains the agent did name as "not reported." The response was logged as `is_synthetic: false`.

Request and response

What the agent requested GPT-5 · matches the <test> in agent_messages

```
Dermatology punch biopsies of the mottled/telangiectatic skin lesions (deep punch to
subcutis) for H&E and direct immunofluorescence; include bacterial/fungal/AFB stains
```

What the Gatekeeper returned GPT-5 · /test_logs/53/result · is_synthetic: false

```
Dermatology deep punch biopsies (livedoid/telangiectatic lesions and clinically normal skin):
Epidermis unremarkable. Numerous small vessels in deep dermis and subcutis (capillaries,
postcapillary venules, small arterioles) are distended/occluded by cohesive aggregates of
large atypical mononuclear cells with scant cytoplasm, vesicular nuclei, prominent nucleoli;
mitoses frequent; occasional intraluminal fibrin thrombi. Minimal perivascular inflammation;
atypical cells confined intraluminally. Similar findings present in biopsy of normal-
appearing skin.
```

Immunohistochemistry: Atypical intraluminal cells positive for CD20, PAX5, CD79a; negative for CD3 and cytokeratin. Endothelial markers (CD31/CD34) highlight endothelium encasing intraluminal aggregates. Cells express MUM1/IRF4 and BCL2; CD10 negative; BCL6 weak; subset CD5 positive. Ki-67 proliferation index approximately 95%. EBER in-situ hybridization: negative. Direct immunofluorescence and special stains for bacteria/fungi/AFB: not reported.

Raw JSON log


```
{
  "test_name": "Test to order: Dermatology punch biopsies of the mottled/telangiectatic skin lesions (deep punch to subcutis) for H&E and direct immunofluorescence; include bacterial/fungal/AFB stains",
  "reason": "",
  "result": "Dermatology deep punch biopsies (livedoid/telangiectatic lesions and clinically normal skin): Epidermis unremarkable. Numerous small vessels in deep dermis and subcutis (capillaries, postcapillary venules, small arterioles) are distended/occluded by cohesive aggregates of large atypical mononuclear cells with scant cytoplasm, vesicular nuclei, prominent nucleoli; mitoses frequent; occasional intraluminal fibrin thrombi. Minimal perivascular inflammation; atypical cells confined intraluminally. Similar findings present in biopsy of normal-appearing skin.\n\nImmunohistochemistry: Atypical intraluminal cells positive for CD20, PAX5, CD79a; negative for CD3 and cytokeratin. Endothelial markers (CD31/CD34) highlight endothelium encasing intraluminal aggregates. Cells express MUM1/IRF4 and BCL2; CD10 negative; BCL6 weak; subset CD5 positive. Ki-67 proliferation index approximately 95%. EBER in-situ hybridization: negative. Direct immunofluorescence and special stains for bacteria/fungi/AFB: not reported.",
  "is_synthetic": false,
  "timestamp": "2026-10-25T18:41:25.721187",
  "batch_mode": true
}
```


An unrequested immunohistochemistry panel was supplied

The agent received B-cell markers (CD20, PAX5, CD79a), a Ki-67 index of about 95% and a negative EBER result. It got diagnostically important information without choosing the detailed panel itself. These findings do exist in the source case report.

Source case report

> Immunohistochemical staining established the lineage and localization of these cells. The atypical intraluminal cells were strongly positive for the B-cell markers CD20, PAX5, and CD79a, and were negative for the T-cell marker CD3 and for cytokeratin, excluding a carcinoma. Endothelial markers, CD31 and CD34, decorated a thin rim of endothelium surrounding the neoplastic aggregates, confirming unequivocally that the malignant cells resided *within* the vascular lumina rather than in the perivascular tissue. The cells expressed MUM1/IRF4 and BCL2 and were negative for CD10, with only weak BCL6 expression, indicating a non–germinal-center (activated B-cell–like) immunophenotype. CD5 was expressed by a subset of cells. The proliferation index, assessed by Ki-67, was very high, at approximately 95%. In-situ hybridization for Epstein–Barr virus–encoded RNA (EBER) was negative.

The tests it did name came back “not reported”

Direct immunofluorescence and the bacterial, fungal and AFB stains were on the order, but the Gatekeeper answered them as “not reported.” What was requested and what was returned do not line up.

The Gatekeeper’s own instructions pull in two directions

The prompt forbids diagnosis-identifying information beyond what is requested, and it also tells the Gatekeeper to return an existing report verbatim. It does not define which ancillary tests belong to a biopsy order. This example alone does not establish a definite rule violation.

Gatekeeper prompt · v6_fixed

> **Prohibited disclosures** – never reveal or imply: … any information *beyond what is requested* that identifies the diagnosis (e.g. "consistent with X" or "suggestive of X")

> If the investigation already exists in the vignette → return its report verbatim.

Why this may affect evaluation

If the rule is to return only explicitly named tests, this exceeds scope. If a biopsy order includes the pathology department’s follow-up panel, it may be allowed, but that scope and its full cost must apply to every participant. This is not a claim that the tests are clinically inappropriate.

CSF cytology also returned flow cytometry `test_logs/52`

Beyond the request | Requested items not reported

Source-backed

The agent ordered a lumbar puncture with opening pressure, cell count, protein and glucose, Gram stain and culture, viral PCRs, cryptococcal antigen, fungal and AFB studies, and, last, CSF cytology. Flow cytometry was not on the list. The response was logged as `is_synthetic: false`.

Request and response

What the agent requested GPT-5 · matches the <test> in agent_messages

```
Lumbar puncture after neuroimaging: opening pressure; CSF cell count with differential, protein, glucose; Gram stain and bacterial culture; HSV-1/2 PCR; VZV PCR; enterovirus PCR; cryptococcal antigen; fungal culture; AFB smear/culture; CSF cytology
```

What the Gatekeeper returned GPT-5 · /test_logs/52/result · is_synthetic: false

```
Lumbar puncture: Opening pressure 17 cm H2O. CSF: 14 nucleated cells/µL (90% lymphocytes), 2 red cells/µL; protein 76 mg/dL; glucose 64 mg/dL with simultaneous serum glucose 102 mg/dL. Gram stain: no organisms; bacterial cultures: no growth. HSV-1/2 PCR: negative; VZV PCR: negative; cryptococcal antigen: negative. CSF cytology: reactive lymphocytes, no malignant cells. CSF flow cytometry: no clonal lymphoid population. Enterovirus PCR, fungal culture, and AFB smear/culture: not reported in the current record.
```

Raw JSON log


```
{
  "test_name": "Test to order: Lumbar puncture after neuroimaging: opening pressure; CSF cell count with differential, protein, glucose; Gram stain and bacterial culture; HSV-1/2 PCR; VZV PCR; enterovirus PCR; cryptococcal antigen; fungal culture; AFB smear/culture; CSF cytology",
  "reason": "",
  "result": "Lumbar puncture: Opening pressure 17 cm H2O. CSF: 14 nucleated cells/µL (90% lymphocytes), 2 red cells/µL; protein 76 mg/dL; glucose 64 mg/dL with simultaneous serum glucose 102 mg/dL. Gram stain: no organisms; bacterial cultures: no growth. HSV-1/2 PCR: negative; VZV PCR: negative; cryptococcal antigen: negative. CSF cytology: reactive lymphocytes, no malignant cells. CSF flow cytometry: no clonal lymphoid population. Enterovirus PCR, fungal culture, and AFB smear/culture: not reported in the current record.",
  "is_synthetic": false,
  "timestamp": "2026-10-25T18:41:25.721185",
  "batch_mode": true
}
```


An unrequested flow cytometry result was added

The Gatekeeper appended `CSF flow cytometry: no clonal lymphoid population`. Cytology and flow cytometry are distinct tests, and only cytology was ordered. The result is in the source case report.

Source case report

> A diagnostic lumbar puncture was performed. The opening pressure was 17 cm of water. The cerebrospinal fluid (CSF) contained 14 nucleated cells per cubic millimeter (90% lymphocytes) and 2 red cells per cubic millimeter; the protein level was 76 mg/dL (reference range, 15–45) and the glucose level was 64 mg/dL with a simultaneous serum glucose of 102 mg/dL. Gram staining and bacterial cultures were negative, as were polymerase-chain-reaction assays for herpes simplex virus and varicella–zoster virus and a cryptococcal antigen assay. CSF cytologic examination revealed reactive lymphocytes but no malignant cells, and flow cytometry of the CSF identified no clonal lymphoid population.

Some requested items came back “not reported”

Enterovirus PCR, fungal culture and AFB smear/culture were on the order but were returned as “not reported in the current record.” None of them appear in the source report. In the same run, the Gatekeeper handled other tests missing from the source by generating results instead (see the two findings below).

Why this may affect evaluation

The agent receives a result for a test it did not select. If the benchmark measures test-level information gathering and cost, what is disclosed should match the actual order. If CSF testing is treated as a package, its components and price should be fixed in advance. As with the skin biopsy, whether this is a violation depends on that package boundary.

Urine culture returned a negative result not in the source `test_logs/29`

Not in source | Source and generated mixed

Synthetic

The agent explicitly ordered a urinalysis and urine culture. It did not write the negative result itself. The response was flagged `is_synthetic: true`.

Request and response

What the agent requested GPT-5 · matches the <test> in agent_messages

```
Urinalysis and urine culture
```

What the Gatekeeper returned GPT-5 · /test_logs/29/result · is_synthetic: true

```
Urinalysis: Specific gravity 1.015; trace protein; no glucose; no leukocyte esterase or
nitrites; 0–2 RBC/HPF; 0–2 WBC/HPF; no casts.
Urine culture: No growth at 48 hours.
```

Raw JSON log


```
{
  "test_name": "Test to order: Urinalysis and urine culture",
  "reason": "",
  "result": "Urinalysis: Specific gravity 1.015; trace protein; no glucose; no leukocyte
esterase or nitrites; 0–2 RBC/HPF; 0–2 WBC/HPF; no casts.\nUrine culture: No growth at 48
hours.",
  "is_synthetic": true,
  "timestamp": "2026-10-25T18:41:25.721160",
  "batch_mode": true
}
```

Source-backed and generated results in one response

The urinalysis findings correspond to the source; the urine-culture result does not. The source's "three sets of blood cultures" are blood cultures, not urine cultures. Because `is_synthetic` is a single flag for the whole response, it does not tell you which line was generated.

Source case report

> The urinalysis showed a specific gravity of 1.015, trace protein, no glucose, no leukocyte esterase or nitrites, and 0–2 red cells and 0–2 white cells per high-power field, without casts. Three sets of blood cultures obtained over 48 hours showed no growth. Testing for human immunodeficiency virus (HIV) antigen and antibody was negative; hepatitis B surface antigen, hepatitis B core antibody, and hepatitis C antibody were negative; and a rapid plasma reagin test was nonreactive. Antinuclear antibody, anti–double-stranded DNA antibody, antineutrophil cytoplasmic antibodies (both proteinase-3 and myeloperoxidase specificities), rheumatoid factor, and anti–cyclic citrullinated peptide antibody were all negative or within normal limits. The serum C3 and C4 concentrations were normal. Cryoglobulins were not detected. A lupus anticoagulant assay was negative, and anticardiolipin and anti–β2-glycoprotein I antibodies (IgG and IgM) were within normal limits. Serum and urine protein electrophoresis with immunofixation showed no monoclonal protein. Serologic testing for *Bartonella henselae*, *Coxiella burnetii*, *Brucella* species, and *Borrelia burgdorferi* was negative, as was a tuberculosis interferon-γ release assay.

The current configuration allows this

For tests absent from the report, the Gatekeeper prompt says to fabricate a plausible synthetic result and flag it. Generation is therefore not evidence of an implementation error. The design question is how the validity and consistency of generated results are established.

▾ Gatekeeper prompt · v6_fixed

> Else → fabricate a plausible synthetic result and set `<is_synthetic>true</is_synthetic>`.

Why this may affect evaluation

A generated negative culture could lead an agent to deprioritize a urinary source of fever. This single run does not establish that it actually changed the model's decision.

▾ **Anaplasma and Ehrlichia returned generated negative PCR and serology** `test_logs/43` Synthetic

Not in source

The agent explicitly ordered Anaplasma and Ehrlichia PCR/serology. The source `full_text` contains no Anaplasma or Ehrlichia entries. The response was flagged `is_synthetic: true`.

Request and response

What the agent requested GPT-5 · matches the <test> in agent_messages

```
Anaplasma and Ehrlichia PCR/serology
```

What the Gatekeeper returned GPT-5 · /test_logs/43/result · is_synthetic: true

```
Anaplasma phagocytophilum:
- PCR: Not detected
- IgG: Negative
- IgM: Negative
Ehrlichia chaffeensis:
- PCR: Not detected
- IgG: Negative
- IgM: Negative
```

Raw JSON log


```
{
  "test_name": "Test to order: Anaplasma and Ehrlichia PCR/serology",
  "reason": "",
  "result": "Anaplasma phagocytophilum:\n- PCR: Not detected\n- IgG: Negative\n- IgM:
Negative\nEhrlichia chaffeensis:\n- PCR: Not detected\n- IgG: Negative\n- IgM: Negative",
  "is_synthetic": true,
  "timestamp": "2026-10-25T18:41:25.721175",
  "batch_mode": true
}
```

Negative PCR, IgG and IgM results were generated

The source lists serologies for Bartonella, Coxiella, Brucella and Borrelia and a tuberculosis interferon-γ release assay, but not Anaplasma or Ehrlichia. The Gatekeeper generated “PCR: Not detected” and negative IgG and IgM for both organisms.

Source case report

> … Serum and urine protein electrophoresis with immunofixation showed no monoclonal protein. Serologic testing for *Bartonella henselae*, *Coxiella burnetii*, *Brucella* species, and *Borrelia burgdorferi* was negative, as was a tuberculosis interferon-γ release assay.

Anaplasma and Ehrlichia appear nowhere in the source `full_text`. The full infectious and autoimmune paragraph is under the urine-culture finding above.

Generated values make the environment variable

Which values get generated for the same request can change the evaluation environment. That mixes Gatekeeper behavior into the score alongside the model being evaluated. This, too, is allowed by the current prompt.

Why this may affect evaluation

Generated negatives could lead an agent to drop tick-borne infections from its differential. This single run does not establish that they changed the model's decision.

<details>
<summary>Benchmark and grading details</summary>

How SDBench is scored

For each case, SDBench runs three roles in a loop. The diagnostic agent receives a short case summary that does not reveal the diagnosis and, turn by turn, orders tests, asks questions or commits to a diagnosis. The Gatekeeper answers each order from the case's full text. When the case contains the result, it returns that; when it does not, its prompt tells it to generate a synthetic result. In this run it sometimes answered "not reported" instead (the first two findings). When the agent commits, a Judge scores the diagnosis against the ground truth on a 1–5 scale; 4–5 counts as correct.

The Gatekeeper and Judge are fixed by the configuration; only the diagnostic agent is being evaluated. So whatever the Gatekeeper discloses feeds straight into the agent's score.

This run

Diagnostic model / agent	GPT-5 / SDBench baseline
Gatekeeper / Judge	GPT-5 / GPT-5 · prompts `v6_fixed` / `v4_reasoning_with_tests`
Case	`synthetic` dataset · train · case 0 — "Case 17-2026: A 58-Year-Old Woman with Recurrent Fever, Progressive Confusion, and Multifocal Cerebral Infarction"

</details>

Action sequence	18 patient questions → 36 test requests in one agent turn → final diagnosis
Final diagnosis	Intravascular large B-cell lymphoma (IVLBCL) with skin and central nervous system involvement. Free-text answer; no multiple-choice options.
Reference diagnosis	Intravascular large B-cell lymphoma (non–germinal-center B-cell immunophenotype), with central-nervous-system and cutaneous involvement and associated secondary hemophagocytic lymphohistiocytosis.
Judge result	4/5, `correct: true`. The Judge noted the omitted non–germinal-center B-cell immunophenotype and the omitted secondary HLH.
Test responses flagged synthetic	10 of 36. The 54-entry log also holds 18 patient-question responses, none flagged.

Reading is_synthetic

`is_synthetic` is a label the Gatekeeper model sets itself; it is not an independent check. `false` means the response was not marked as containing fabricated results. It does not certify accuracy, appropriate scope or real-patient provenance. `true` means generated content is included, not necessarily that the whole response was generated. The source report is itself synthetic.

The label on the right of each finding combines this flag with our own check. **Synthetic** means the response was flagged `true`. **Source-backed** means it was flagged `false` and we confirmed its findings appear in the source report, apart from the lines answered “not reported.”

What this log supports

This is not evidence that the diagnostic model assumed normal results without ordering tests. The model requested tests; the Gatekeeper supplied extra information or generated missing results. This log alone cannot establish how much that affected the final diagnosis.

The core diagnosis was correct

The model answered IVLBCL after receiving the skin-pathology findings. The log does not support calling this an unsupported guess.

The test-selection process was not validated

Results beyond the explicitly named tests were supplied, and the agent submitted all 36 test requests in one turn.

The Gatekeeper may influence evaluation

The log shows expanded disclosure and generation of missing results. Establishing a causal effect on the final answer requires a separate experiment.

A token-saving motive cannot be established

The agent prompt sets no token objective, and the log records no per-test rationale: the harness never fills the `reason` field, so it is empty for every entry. We do not infer the model's intent.

Conditions for a consistent comparison

1. Define the returned findings and included follow-up panels for each test ID.
2. Align the test components disclosed to the agent with those included in costing.
3. Return "unavailable" for missing results, or use reviewed, fixed values shared across participants.
4. Record source-backed versus generated provenance for each test component, not only for the combined response.

Sources and reproducibility

- Trajectory: `results/seungju_synthetic_baseline_gpt-5_train_20261025_183616_k1/json/case_k0_0.json`
- Source report: `data/processed_case_data/synthetic/case_0_Case_17_2026__A_58_Year_Old_Woman_with_Recurrent_F.json`
- Run configuration: `run_config.json` · Gatekeeper prompt: `sdbench/prompts/gatekeeper_synthetic_tests/v6_fixed/prompt.yaml`
- Trajectory SHA-256: `19cdc9815b028e97db4fa9609f12fde44c670761079e84806f302f0ecceb1222`

This is an information-flow review of a saved run. It is not an expert clinical assessment, a controlled comparison with human physicians, or a rerun with selected information removed. The original diagnosis and cost results are unchanged.

Want the full evidence?

Full trajectory audit ↗